🧠 SEIZURE DISORDERS

🔍 WHAT ARE SEIZURES?

- Caused by **abnormal, rapid electrical firing** of neurons (nerve cells that transmit electrical signals) in the brain
- Result from **disruption between excitatory (stimulating) and inhibitory (calming)** neurons
- **Types of neurons:**
  - **Excitatory neurons** → release **glutamate** (a chemical that stimulates brain activity)
  - **Inhibitory neurons** → release **GABA** (gamma-aminobutyric acid, a chemical that calms brain activity)

📊 CLASSIFICATION OF SEIZURES

✅ GENERALIZED SEIZURES (AFFECTING BOTH SIDES OF THE BRAIN)

1. Tonic-Clonic (Grand Mal) ("tonic" = stiffening; "clonic" = jerking; "grand mal" = older term for severe seizure)

- Aura (early warning sign) may be present
- Tonic phase: Body stiffens, apnea (absence of breathing), cyanosis (bluish skin from lack of oxygen), biting of tongue
- Clonic phase: Jerking movements, possible incontinence (loss of bladder/bowel control)
- High risk for **Status Epilepticus** (seizure lasting >5 minutes or multiple without recovery)
- Postictal (recovery phase): Confused, tired, sore

2. Absence (Petit Mal) ("petit mal" = mild seizure, old term)

- Common in children
- Sudden staring episodes
- Very brief (seconds), with immediate postictal recovery
- May go unnoticed

Atonic (Drop Attacks) ("a" = without; "tonic" = muscle tone → sudden limpness)

- Sudden loss of muscle tone → falls
- High risk for **head injury**
- Often need helmets
- Immediate postictal recovery

Tonic-only or **Clonic-only**

- Tonic: Stiffening only
- Clonic: Jerking only

✅ FOCAL (PARTIAL) SEIZURES (AFFECTING ONE AREA OF THE BRAIN)

1. Focal Onset Aware (Simple Partial)

- Patient remains **aware**
- Symptoms depend on location (e.g. occipital = visual changes)
- Often serves as the **aura** phase (early symptom before bigger seizure)

2. Focal Impaired Awareness (Complex Partial)

- Patient is **unaware** during seizure
- May display **automatisms** (involuntary, repetitive movements like lip-smacking)
- Most common in **temporal lobe** (part of brain related to memory and emotion)

⏳ STAGES OF A SEIZURE

1. Prodromal – Days before seizure

- Depression, anxiety, GI symptoms (gastrointestinal issues like nausea, cramps), urinary changes

2. Aura – Seconds to minutes before seizure

- Only occurs in some types (e.g. tonic-clonic, focal)
- Patient may experience smells, sounds, vision changes, déjà vu (feeling of familiarity), dizziness

3. **Ictus** – The seizure itself

- Time it!
- Lasts **1–3 minutes** typically
- **>5 minutes = STATUS EPILEPTICUS** (emergency condition of prolonged seizure)

4. **Postictal** – Recovery phase

- Fatigue, confusion, headache, amnesia (memory loss)
- Duration depends on seizure type

🧪 COMMON CAUSES OF SEIZURES

- Fever (especially in children)
- Hypoglycemia (low blood sugar)
- CNS infections (central nervous system, e.g., meningitis)
- Alcohol withdrawal
- Brain tumors
- Acid-base imbalance (e.g., acidosis = excess acid in body fluids)
- Hypoxia (low oxygen in tissues)
- Stroke
- Traumatic brain injury
- Congenital conditions (present from birth)
- Idiopathic (unknown cause)

🚑 NURSING INTERVENTIONS

🔒 SEIZURE PRECAUTIONS

- Pad side rails
- Suction and oxygen set up at bedside
- IV access secured
- Bed in lowest position
- Remove restrictive clothing/objects
- Know the patient's seizure history (aura, triggers, typical duration)

🔄 DURING A SEIZURE

- **Do NOT restrain**
- **Do NOT insert anything in the mouth**
- Turn patient to **side** (protect airway)
- Place **pillow under head**
- Time the seizure!
- Remove glasses and hazardous items

🧘 AFTER A SEIZURE

- Assess airway, vitals, neuro status
- Allow rest
- Document seizure characteristics (duration, movements, responsiveness)
- Notify provider
- Prepare for labs, medications, or EEG (electroencephalogram — brain wave test)

⚠️ HIGH-YIELD: STATUS EPILEPTICUS

- Seizure lasting >**5 minutes** or **recurrent seizures without recovery**
- **Emergency** → activate response, give benzodiazepines (fast-acting anti-seizure drugs)

🧠 DIAGNOSTIC TEST: EEG (ELECTROENCEPHALOGRAM)

- Measures electrical brain activity
- No caffeine 8 hrs prior
- Hold seizure meds (if ordered)
- Clean, dry hair (no oils/products)

💊 COMMON ANTI-SEIZURE MEDICATIONS

Drug Class	Example	Key Points
Barbiturates	Phenobarbital	Stimulates GABA (inhibitory neurotransmitter), watch for sedation, hypotension (low blood pressure), **therapeutic level: 15–40**
Hydantoins	Phenytoin (Dilantin)	Causes **gingival hyperplasia** (gum overgrowth), rash, avoid milk/antacids, **therapeutic level: 10–20**
Benzodiazepines	Lorazepam, Diazepam	Used in **status epilepticus**, reversal agent: **Flumazenil** (antidote for overdose)
Valproates	Valproic Acid	Monitor **liver function**, platelets, GI upset (stomach discomfort)

🧠 OTHER TREATMENTS

- **Surgery** (e.g. **temporal lobectomy** = removing part of the temporal lobe)
- **Vagus Nerve Stimulator** (implanted device that sends electrical pulses to reduce seizures)
- **Ketogenic Diet** – High-fat, low-carb (esp. in children with uncontrolled epilepsy)

🔑 MNEMONIC: "STOP SEIZURE" – TRIGGERS TO AVOID

Letter	Trigger	Explanation
S	Stress	Mental/emotional strain
T	Trauma	Physical injury, esp. head trauma
O	Overexertion	Excessive physical activity
P	Period (hormonal changes)	Menstrual or hormonal fluctuations
S	Sleep loss	Fatigue lowers seizure threshold

Letter	Trigger	Explanation
E	Electrolyte/metabolic issues	Low sodium, calcium, or blood sugar
I	Illness	Infections or fever
Z	sZtimuli (visual, auditory)	Lights, sounds, smells triggering brain activity
U	Under-medicated	Missed doses, non-compliance
R	Recreational drugs	Cocaine, ecstasy, marijuana
E	Ethanol (alcohol)	Alcohol use or withdrawal